ONLINE MASTERS IN **DATA SCIENCE**

DSC258: NATURAL LANGUAGE PROCESSING

NER AND MASSIVE RAW TEXTS

UC San Diego

COMPUTER SCIENCE & ENGINEERING

HALICIOĞLU DATA SCIENCE INSTITUTE

OUTLINE

Classic Named Entity Recognition (NER)

Some questions to think about:

- **Can massive raw texts help?**
  - Can dictionaries help?
  - Are human annotations always correct?
  - Is Tokenizer always good?

Using **Language Model** for better representations:

Word-level Language Model:

- ELMo (Peters et al., NAACL’18, **best paper**)
- LD-Net (Liu et al., EMNLP’18)

Char-level Language Model:

- LM-LSTM-CRF (Liu et al., AAAI’ 18)
- Flair (Akbik et al., COLING’18)

Hybrid Language Model:

- Cross View Training (Clark et al., EMNLP’ 2018)
- BERT (Devlin et al., NAACL’19, **best paper**)

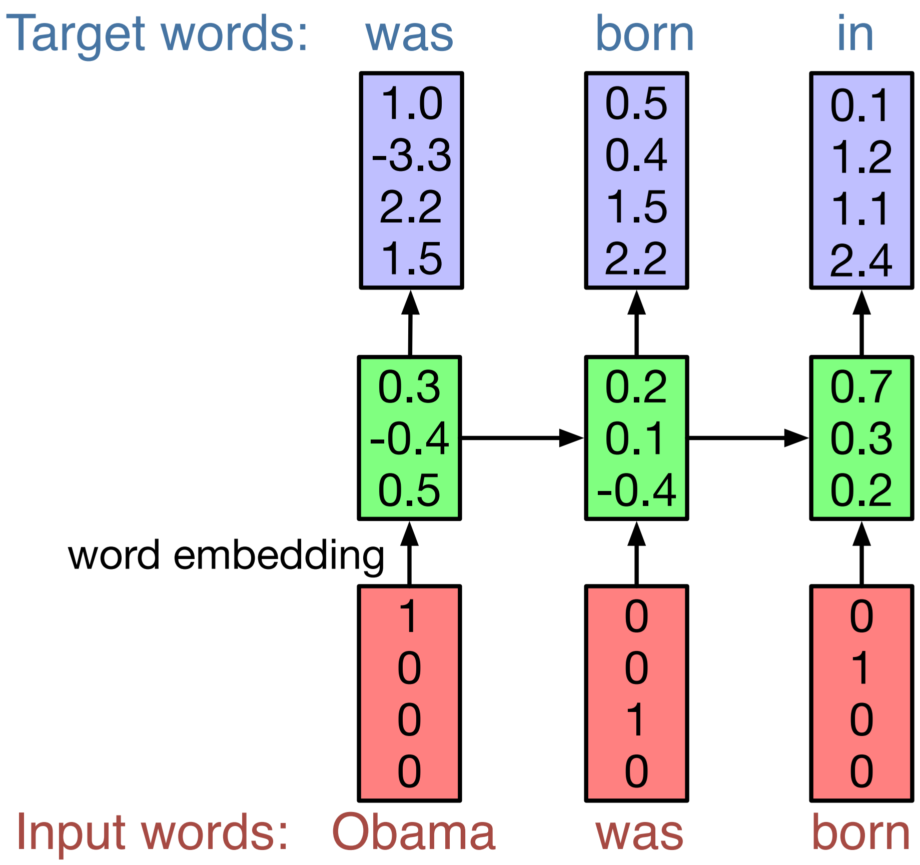
WHAT'S (NEURAL) LANGUAGE MODEL ? (RECAP)

Describing the generation of text:

- predicting the next word based on previous contexts

Pros:

- Does not require any human annotations
- Nearly unlimited training data!
- Resulting models can generate sentences of an unexpectedly high quality



Char-by-Char Markdown Generations:

```
'''See also''': [[List of ethical consent processing]]

== See also ==
*[[lender dome of the ED]]
*[[Anti-autism]]

===[[Religion|Religion]]===
*[[French Writings]]
*[[Maria]]
*[[Revelation]]
```

Valid Syntax!

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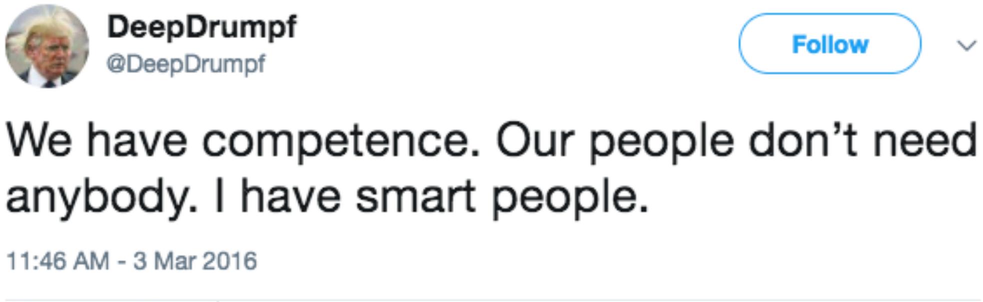
Valid Syntax!

NEURAL LM: EXAMPLE GENERATIONS (RECAP)

Deep “Donald Trump”: Mimics President Trump



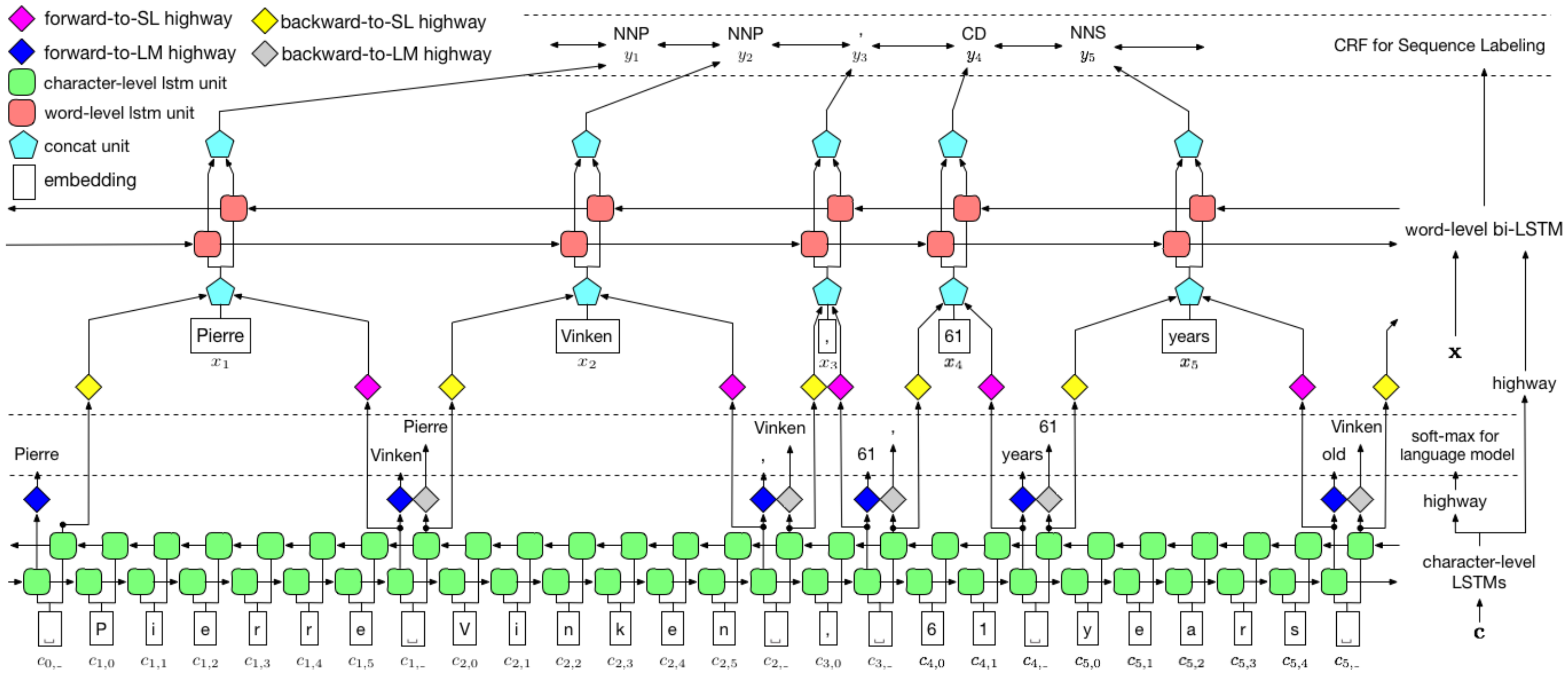
Fooled many twitter users

LM-LSTM-CRF: CO-TRAIN NEURAL LM

Propose to use **Character-level language model** as a **Co-Training** objective

Why character-level?

- More efficient & More robust to pre-processing

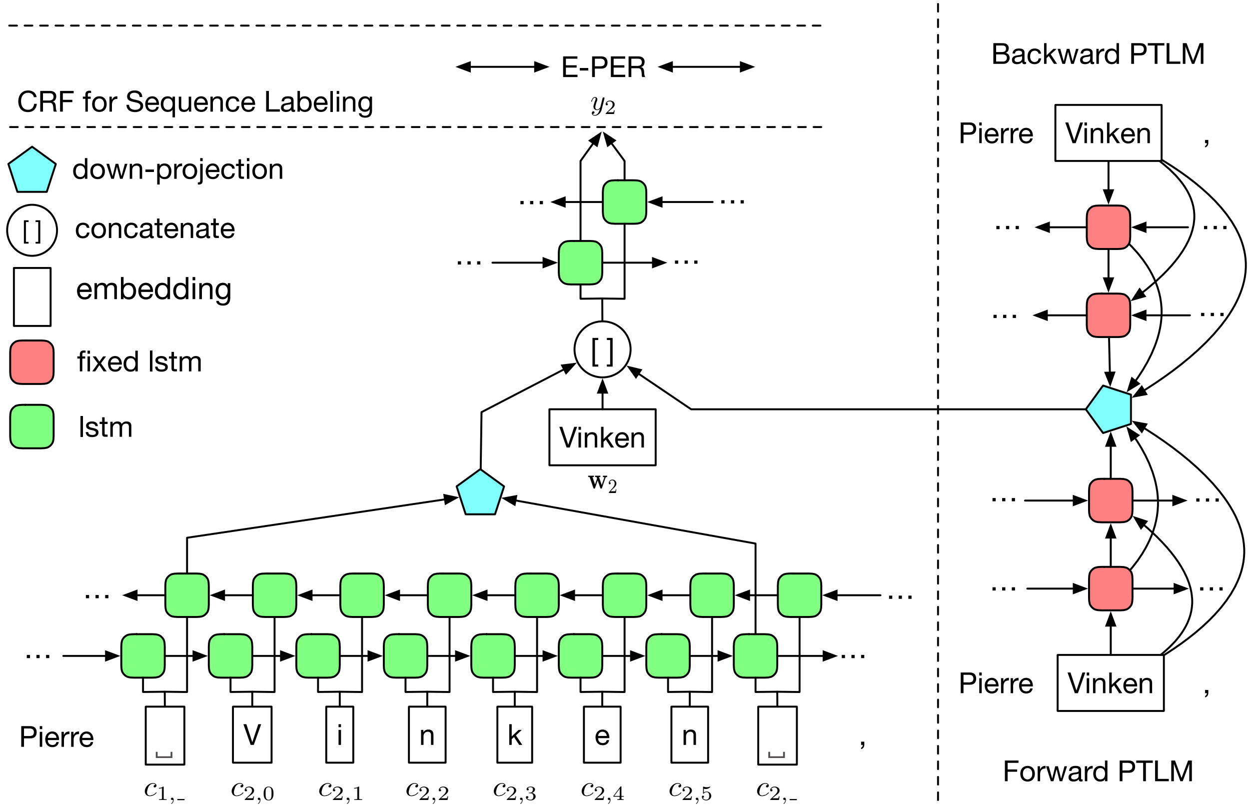


ELMO: PRE-TRAIN WORD-LEVEL NEURAL LM

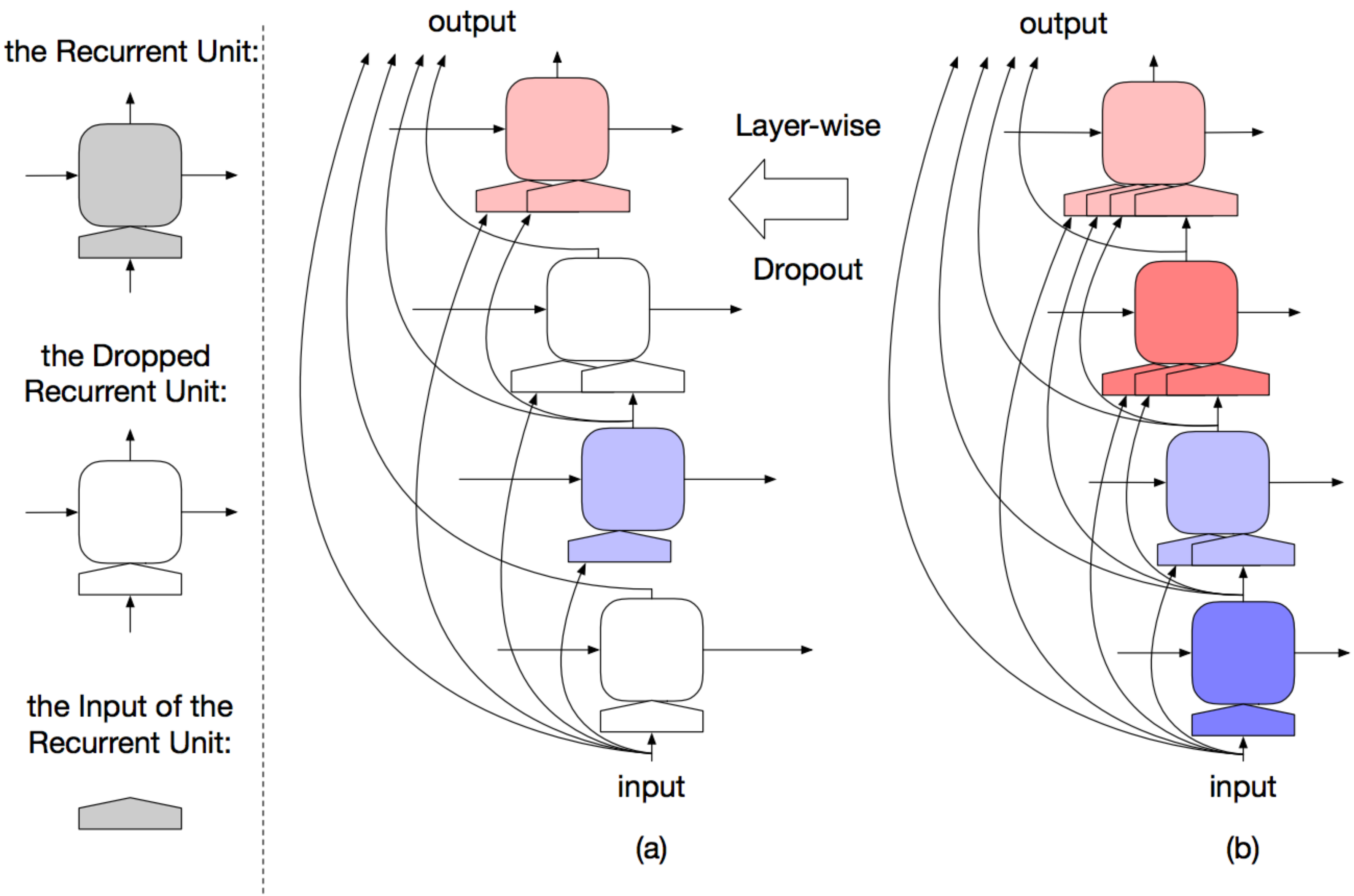
Add ELMo at the input of RNN. For some tasks (SNLI, SQuAD), including ELMo at the output brings further improvements

Key points:

- **Freeze** the weight of the biLM
- Regularization are necessary



LD-NET: AN EFFICIENT VERSION OF ELMO



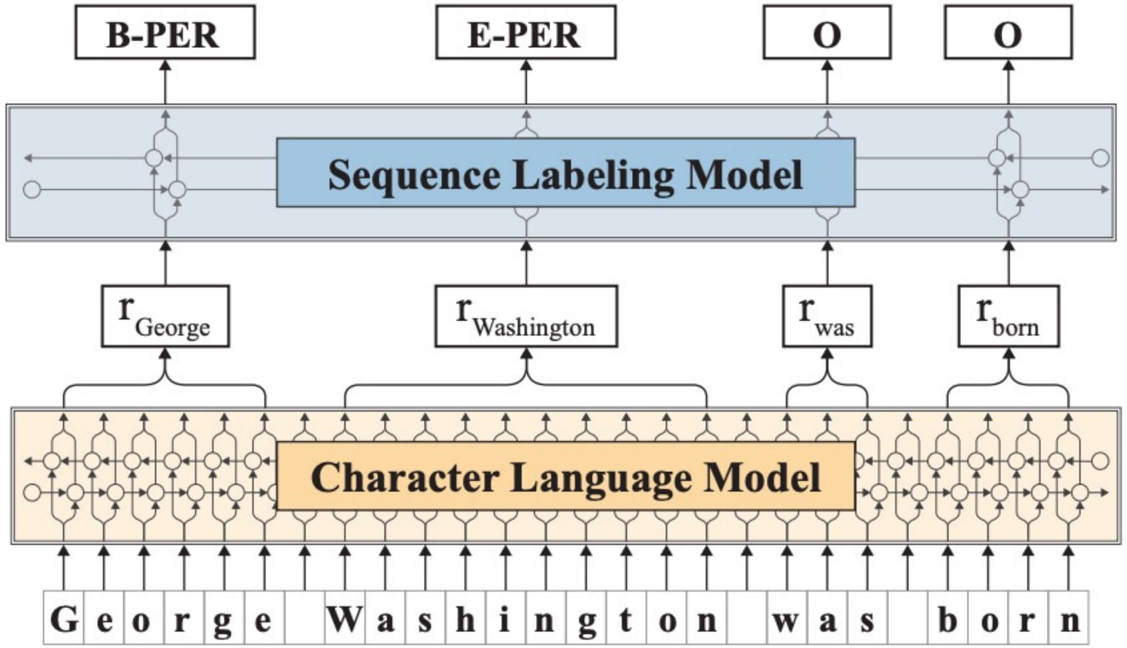
Make the contextualized represent ***efficient without much loss of effectiveness***

Network (PTLMs Ind.#)	Avg. ppl	#FLOPs ($\cdot 10^6$)	F_1 score (avg$\pm$std)
NoLM (/)	/	3	90.78$\pm$0.24
O-ELMo (3)	39.70	607†	92.22$\pm$0.10
R-ELMo (6)	40.27	215	91.99$\pm$0.24
R-ELMo (7)	48.85	135	91.54$\pm$0.10
TagLM (5)	47.50	87†	91.62$\pm$0.23
LD-Net (8)	45.14	98	91.76$\pm$0.18
LD-Net (9)	50.06	98	91.86$\pm$0.15
LD-Net (8*)	origin	98	91.95
	pruned	6	91.55$\pm$0.06
LD-Net (9*)	origin	98	92.03
	pruned	6	91.84$\pm$0.14

FLAIR: PRE-TRAIN NEURAL LM AT ALL LEVELS

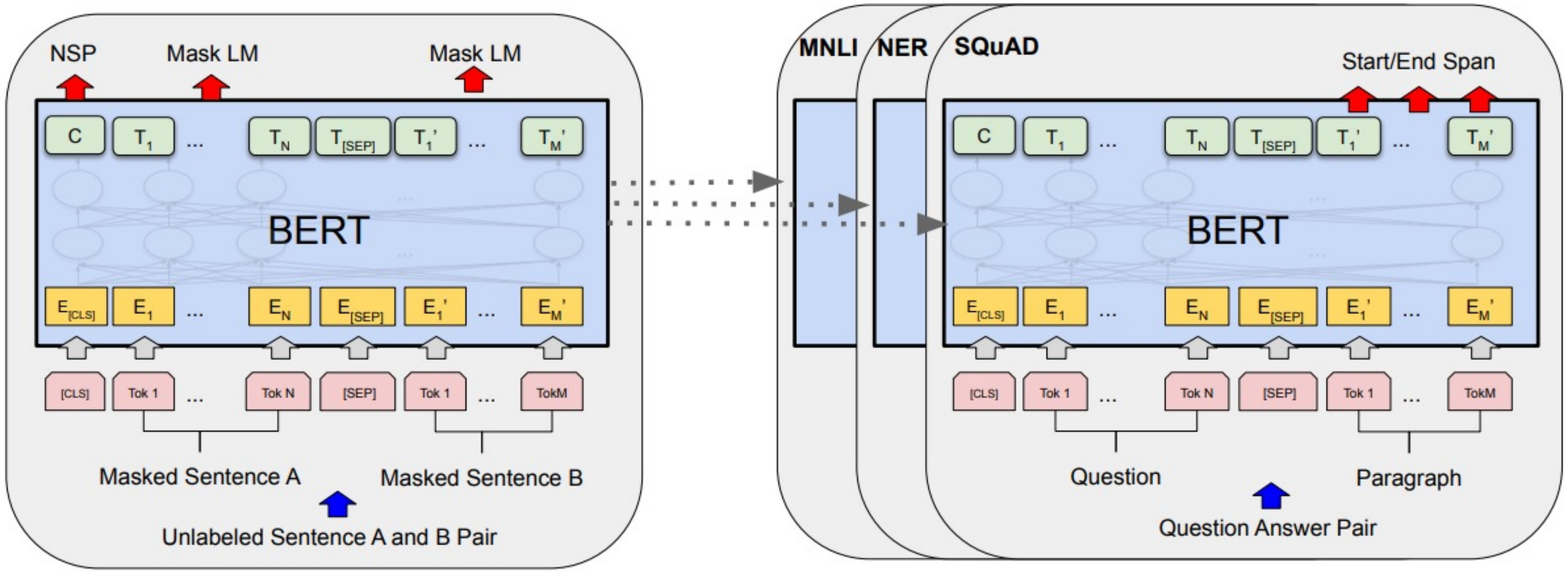
- Even for character-level language model, pre-training is very important.
- The structure is the same with LM-LSTM-CRF, the difference is the pre-training conducted on additional training corpus.

Task	PROPOSED	Previous best
NER English	**93.09**±0.12	*92.22*±0.1 (Peters et al., 2018)
NER German	**88.32**±0.2	*78.76* (Lample et al., 2016)
Chunking	**96.72**±0.05	*96.37*±0.05 (Peters et al., 2017)
PoS tagging	**97.85**±0.01	*97.64* (Choi, 2016)



BERT: INTRODUCE TRANSFORMER

- Introduce Transformers, use masked language model + next sentence prediction
- Conduct fine-tuning after pre-training on each task (necessary for sentence-level tasks, NER is a word level task).



Pre-training Fine-Tuning

System	Dev F1	Test F1
ELMo (Peters et al., 2018a)	95.7	92.2
CVT (Clark et al., 2018)	-	92.6
CSE (Akbik et al., 2018)	-	**93.1**
Fine-tuning approach		
BERT$_{\text{LARGE}}$	96.6	92.8
BERT$_{\text{BASE}}$	96.4	92.4
Feature-based approach (BERT$_{\text{BASE}}$)		
Embeddings	91.0	-
Second-to-Last Hidden	95.6	-
Last Hidden	94.9	-
Weighted Sum Last Four Hidden	95.9	-
Concat Last Four Hidden	96.1	-
Weighted Sum All 12 Layers	95.5	-

Table 7: CoNLL-2003 Named Entity Recognition results. Hyperparameters were selected using the Dev set. The reported Dev and Test scores are averaged over 5 random restarts using those hyperparameters.

NEW STATE-OF-THE-ARTS

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